Fundamental Groups and Covering Spaces

MSU Summer Topology Program

Adithya Sathyanarayana

May 11, 2026

The Motivation for Algebraic Topology



Figure 1: Ceramic model by Keenan Crane and Henry Segerman

Topology, often described as "rubber-sheet geometry", is a major branch of mathematics concerned with the properties of a space that remain unchanged under continuous deformations—such as stretching, twisting, crumpling, and bending, but not tearing or gluing.

Topology focuses on the intrinsic, qualitative structure of a space rather than strict quantitative measurements like distance, angle, or volume.

Homeomorphism

- This is the foundational equivalence in topology. Two spaces are considered topologically identical (**homeomorphic**) if one can be continuously deformed into the other.

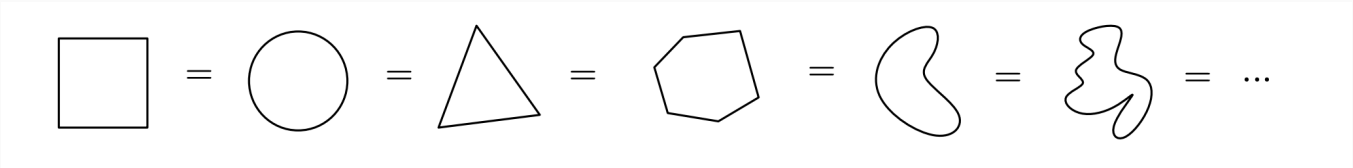


Figure 2: The 'class' of circles

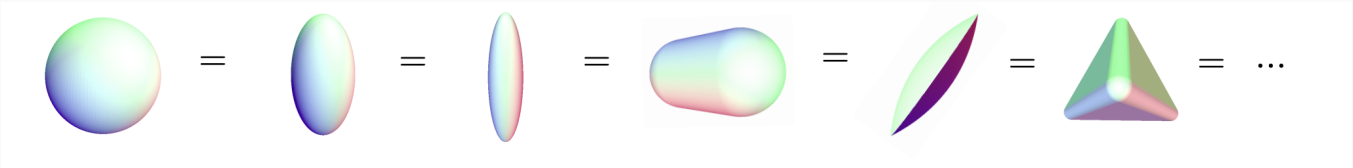
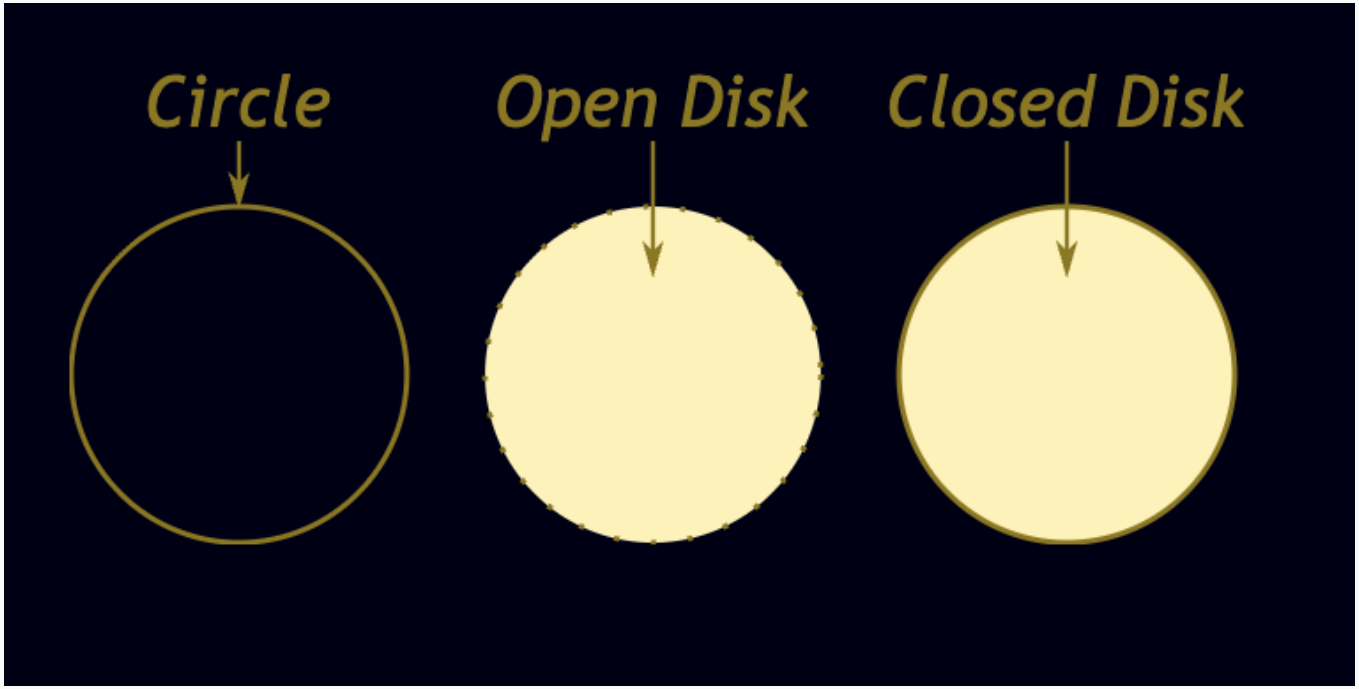


Figure 3: The 'class' of spheres[1]

[1] Image credits: Raphaël Tinarrage

Topological invariants

- We observe specific properties that survive continuous deformation and characterize them as **"topological invariants"**. If two spaces have different invariants, they cannot be homeomorphic.
- Some of these invariants are Connectedness, **Compactness**, Number of Holes.



Topological Invariants



Figure 4: The 'number of holes'[2] invariant

[2]To make this invariant more precise, we use a tool called Homotopy

Homotopy

- A **homotopy** is a *family of continuous maps* $f_t : X \to Y$ that represents a continuous way of deforming X into Y

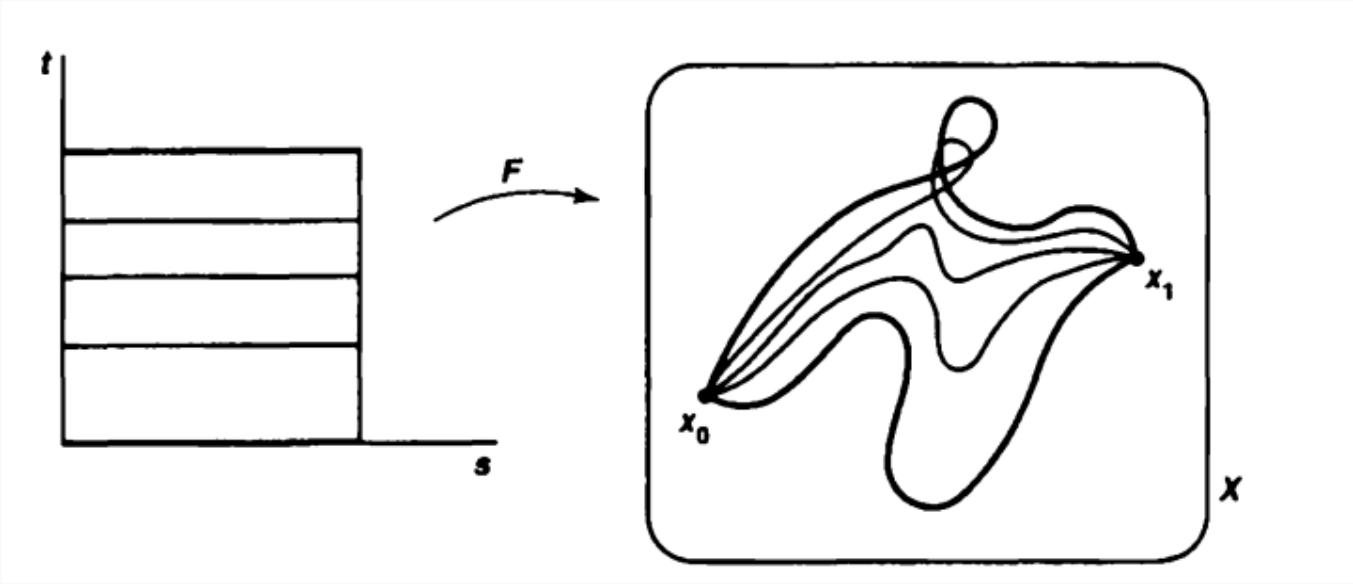
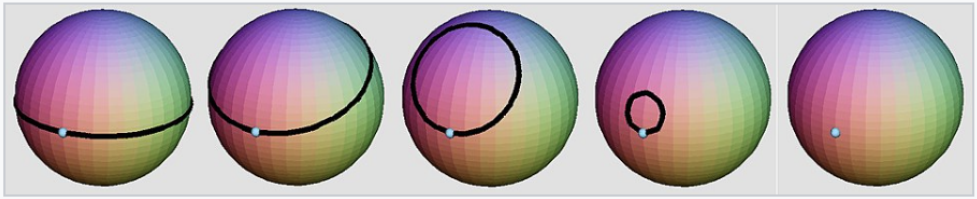


Figure 5: Path[3] homotopy, credits: Munkres textbook

[3] A path f from x_0 to x_1 in X is a continuous map $f : [0, 1] \to X$ such that $f(0) = x_0$ and $f(1) = x_1$

The homotopy of ‘ropes’ analogy to detect a hole



A loop on a 2-sphere (the surface of a ball) being contracted to a point

[4]

[4] Credits: Wikipedia

The Fundamental Group $\pi_1(X, x_0)$

- Let X be a space; let x_0 be a point of X. A path in X that begins and ends at x_0 is called a **loop** based at x_0.
- The **fundamental group** is the *set of homotopy classes of loops* that start and end at a specific basepoint x_0.

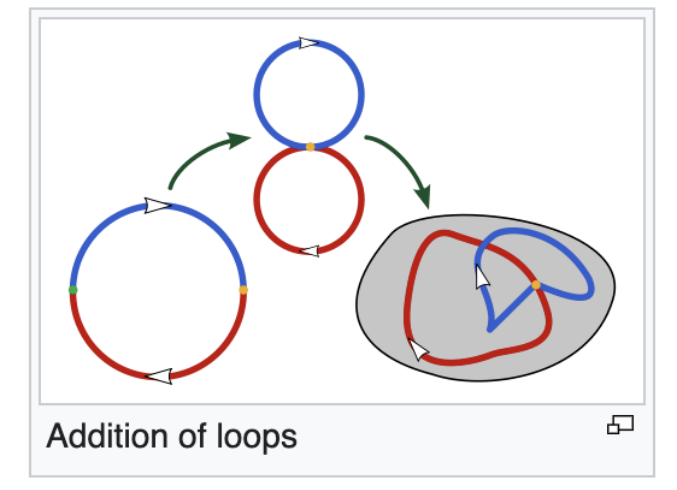
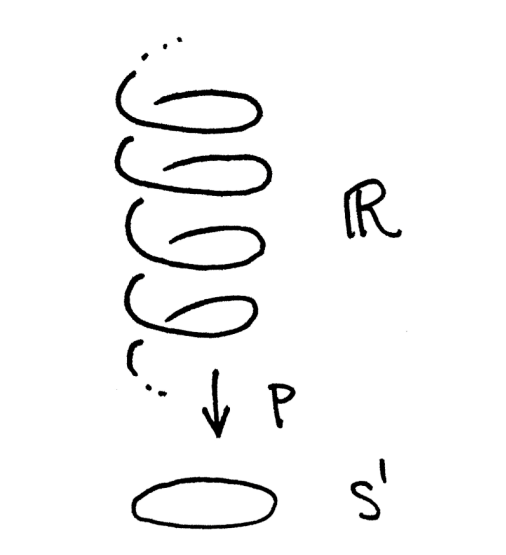


Figure 6: Concatenation of loops, Credits: Wikipedia

Example: The Circle S^1

- The fundamental group of the circle is the **infinite cyclic group** $\mathbb{Z}$.
- **Theorem:** The map $\Phi : \mathbb{Z} \to \pi_1(S^1)$ given by $n \mapsto [\omega_n]$, where $\omega_n(s) = (\cos 2\pi ns, \sin 2\pi ns)$ for $s \in [0, 1]$ is a loop in S^1 based at $(1, 0)$, is an isomorphism.
- Idea: Compare loops in S^1 with loops in (the helix) $\mathbb{R}$



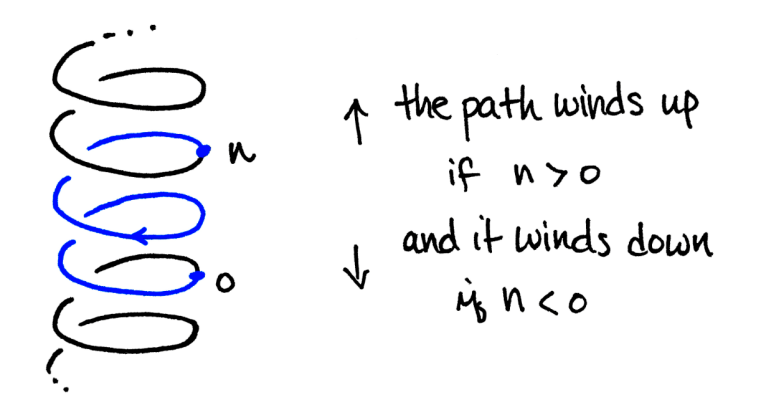


Figure 7: Credits: Math3ma

Covering Spaces

- A useful tool to compute the fundamental groups is the idea of a '**covering space**'.
- A covering space consists of a "total space" E, a "base space" X, and a continuous, surjective map $p : E \to X$ (the covering map)such that every point x in the base space X must have an *evenly covered neighborhood*.

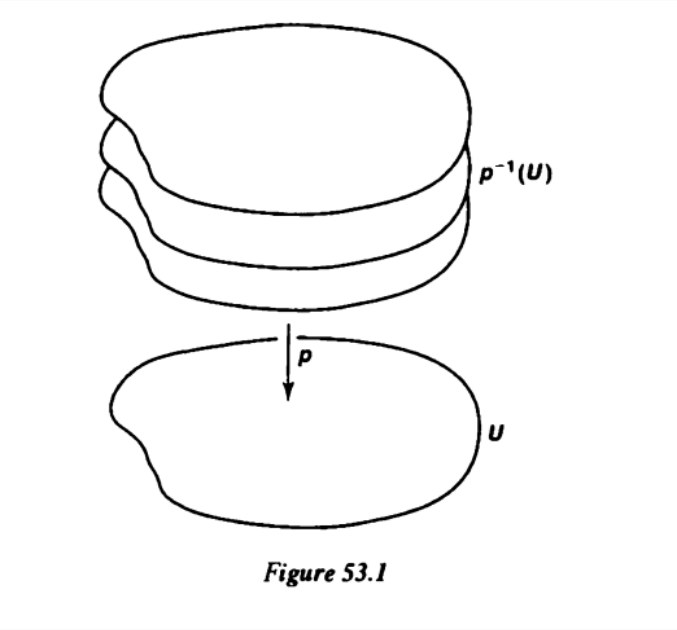


Figure 8: Stack of pancakes, Credits: Munkres

- The projection map $p : \mathbb{R} \to S^1$ we previously discussed is an example of a covering space
- There can be many covering spaces for a single base space.
- However, the real line $\mathbb{R}$ is special because it is simply connected (it has no holes, meaning its fundamental group is trivial). A covering space that is simply connected is called the **Universal Cover**.

- A big theorem in algebraic topology states that the fundamental group of a "nice" space X acts freely on the fiber of its universal cover. Because the universal cover of S^1 is $\mathbb{R}$, and the fiber over a point in S^1 is the integers $\mathbb{Z}$, this gives us proof that the fundamental group of the circle is isomorphic to the integers:

$$\pi_1(S^1) \cong \mathbb{Z}$$